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Smart People for Smart Cities: A Behavioral Framework for Personality and Roles

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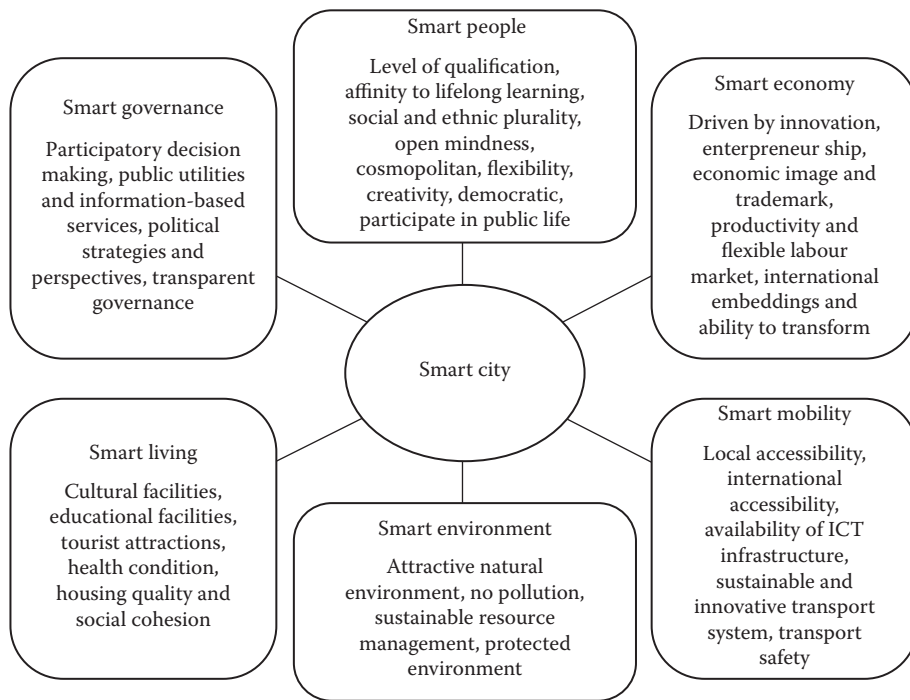
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3.1 Introduction

Creative, sustainable, and livable are some of the key features of the smart city. Developing a smart city may involve various domain experts such as consultants, corporations, marketing specialists, and city officials to frame how the smart city could be conceptualized, understood, and planned.

A smart city may be considered as an assemblage of technologies such as information and communication technologies (ICT), infrastructure, smart transport, and e-governance to increase competitiveness and administrative efficiency, as well as social inclusion.

Since 2011, contributions to the smart city have been critically scrutinized from different viewpoints, such as science and technology, politics, economy, government mentality studies, and ideological critique. In general, smart cities create a relation between technology and society. This process is related to the actor-network theory (ANT), which focuses on the making of sociotechnical networks and how certain actors try to create for their interest.

**FIGURE 3.1**

Dimensions of the smart city. (Adapted from Kumar, T. V. *E-governance for Smart Cities*, Springer, Singapore, 2015.)

3.1.1 Overview of Smart Cities

Various theoretical and marketing conceptualizations have been introduced, such as techno cities (Downey and McGuigan, 1999), creative cities (Florida, 2005), cyber cities (Graham and Marvin, 1999), digital cities (Komninos, 2008), sentient cities (Shepard, 2011), and networked cities (Castells, 1996).

The concept of the smart city originated from various definitions, such as the information city, the intelligent city, the digital city, the knowledge city, and the ubiquitous city. Smart cities represent a conceptual urban development model of human resource utilization and usage of technological capabilities (Angelidou, 2015).

The major dimensions of the smart city are smart economy, smart people, smart environment, smart living, smart mobility, and smart governance (Kumar, 2015). These dimensions are shown in [Figure 3.1](#).

3.1.1.1 Smart Economy

Smart economy is driven by innovation and supported by universities, which provide an ecosystem to grow the entrepreneurial spirit of the people in a society. The economy of the city is such that it creates an economic image globally, and will also have a trademark. Productivity and a flexible labor market should be provided by the city administrative authority. The economy should have international branding and generate a highly diversified economy.

3.1.1.2 Smart Mobility

Smart mobility means that the city should have national and international accessibility, using information and communication technologies (ICTs) to ensure that technology has been used widely during the design of national highways and bridges. Metros, monorails, and an intelligent transport system should be used for daily commuters. The urban design should be such that it gives importance to last-mile connectivity. The transport system should be sustainable and innovative. It should also take care of the safety of daily and occasional commuters within and outside the city.

3.1.1.3 Smart Environment

Smart environment refers to an attractive natural environment with no pollution. Importance should be given to the carbon footprint and the natural resources available. It should have greenery in almost every part of the city. It should have waste management systems as well as natural resource management. The city should also have systems to protect the existing greenery of the city from any external factor.

3.1.1.4 Smart Living

Smart living is characterized by diverse cultural facilities available to all kinds of religions, whether they belong to major or minor communities. Education facilities should be provided by establishing world-class colleges and universities. It should also have tourist attractions as well as world-class hospitals with all the latest technology-enabled devices and equipment to allow a healthy lifestyle for every resident. Good-quality housing as well as social cohesion should be provided to citizens of the city.

3.1.1.5 Smart Governance

Smart governance is an advanced form of e-governance. It is about making the right policy choices and the implementation of same. The knowledge society should give a set of elements for smart governance that is open to sharing information and its use, transparent decision making, collaboration and participation of stakeholders, improvement in government services, and operation through the use of intelligent technologies.

3.1.1.6 Smart People

The Human Development Index is considered to be the most important aspect. The next most important attribute is the graduate enrolment ratio. The third most important attribute is the level of qualification. Smart people should have a lifelong zeal to learn, and there should be social and ethnic plurality. Open-mindedness is another quality of smart people, as is having the flexibility to adapt to changes in the environment, as well as the creativity to contribute to education. Smart people possess a democratic nature and participate in public life.

3.2 Literature Review

The big-five factor representation was introduced by Tupes and Christal in 1961. This was on the basis of a reanalysis of various datasets using bipolar variables given by Cattell in

1957. The five factors are listed as (1) agreeableness; (2) surgency (or extroversion); (3) emotional stability (versus neuroticism); (4) consciousness (or dependability); and (5) culture, intellect, or openness. Norman (1963) selected four variables after the analysis of Tupes and Christal (1961). These four factors have the highest factor loading on each of the five factors. These 20 variables have been used as big-five markers in later studies carried out by Norman, as well as Guthrie and Bennett (1971). Since the variables have some inherent limitation as markers of the big-five structure, only four variables have been included in each of the domains. The Neuroticism-Extraversion-Openness Personality Inventory (NEO-PI) developed by Costa et al. (1991) was the major alternative set of big-five markers. It provides scales to measure five domains, such as neuroticism extraversion (factor I), agreeableness (factor II), conscientiousness (factor III), emotional stability (factor IV), and openness (factor V).

3.3 Research Methodology

Personality traits describe an individual's behavior, such as the way a person interacts with others and reacts to any situation. A person could be evaluated on a series of factors. Factors such as the five-factor model (FFM), which is also known as the big-five personality traits, cover all the characteristics of an individual and explain the major significant variations in human personality. These five factors are extroversion, conscientiousness, openness to experience, agreeableness, and emotional stability, as shown in [Table 3.1](#).

These attributes are powerful predictors of people's behavior. In this chapter, the relationships between personality dimensions and job performance have been studied. The findings show how people should behave and work smartly in a smart city in order to live a high-quality and luxurious life.

3.4 Findings and Discussion

Previous research has shown that individuals who are reliable, responsible, good planners, hardworking, organized, persistent, and have positive attitudes tend to have higher job performance in most occupations. Those with higher levels of job knowledge and problem-solving skills perform better in their jobs. Level of qualification, affinity to lifelong

TABLE 3.1

Five Personality Traits

Factor	Explanation
Extroversion	Shows the sociable, gregarious, and assertive nature of a person
Agreeableness	Describes someone who is cooperative, good-natured, and trusting
Conscientiousness	Indicates persistent, dependable, responsible, and organized nature of a person
Emotional stability	Shows self-confident, calm, secure (positive) versus depressed, nervous, and insecure (negative) traits of a person
Openness to experience	Characterizes someone in terms of sensitivity, curiosity, and imagination

learning, creativity, flexibility, participation in public life, and good decision making are some essential qualities that make people smart (Giffinger et al., 2007) and increase their work performance.

The extroversion dimension captures people's comfort level with relationships and social life. Extroverts tend to be social and assertive, enjoy the company of others, live a happier life, have more positive emotions, perform better in all types of work, are willing to make changes, and participate actively in decision processes. The people in a smart city should be extroverts rather than introverts, who are usually reserved, timid, and quiet.

The agreeableness dimension refers to highly agreeable people who are cooperative, warm, happy, romantic, organizational team members, and are more satisfied in their jobs. This personality trait is required for interpersonally oriented jobs such as customer service. Superior organizational ability leads to greater efficiency in an employee, an artist, a student, a business owner, or a celebrity. People with this attribute may contribute to the enhancement of the living environment of a smart city by engaging citizens in various city activities.

The conscientiousness dimension is a measure of reliability. A highly conscientious person is less distracted and more organized, responsible, dependable, interested in new learnings, and persistent. These types of people generally live longer since they take care of themselves in better ways. Paying attention to what happens in their surroundings and acting accordingly are key factors for the success of smart people in a particular smart city. This personality dimension can create a safe, secure, corruption-free, crime-free, and healthy environment in a smart city.

The emotional stability dimension is mostly related to job satisfaction, life satisfaction, and low stress levels. Such people are calm, self-confident, optimistic, and not vulnerable to the psychological and physical effects of stress. Hope for the best possible scenario and preparing for the worst indicates the far vision of smart people. People with good emotional stability have a better way of overcoming their obstacles and finding the solutions to compensate for their shortcomings. This dimension of personality can lead to a frustration-free life environment in a smart city.

Extremely open people are curious, creative, and artistically sensitive. Creativity is important to leadership, as open people are more likely to be effective leaders, and more comfortable with change and ambiguity. This quality makes people compatible when exposed to a new environment or experience. The situation could be controlled and negative impacts on lives could be prevented if a person learns from his or her mistakes. Smart people always view a situation from every possible angle in order to find the best solutions. People with this personality trait don't lose the opportunity to broaden their experience. Openness may provide good acting leaders who identify new opportunities in a smart city and also participate in order to change situations according to the citizens' needs, future growth, and prosperity. [Table 3.2](#) shows the relationships among the personality traits, people performance, quality of smart people, and the roles of people in a smart city's development.

Smart people understand the importance of people in their surroundings to get support, opportunities, and to make a healthy environment. Smart people can generate a pleasant city environment and a positive workforce. Making smart decisions in different situations leads people to face reality and judge themselves honestly to overcome problems. Being smart is not just about intelligence, critical thinking, logical reasoning, and scientific approaches, but it is also about constantly working to improve the capability for various tasks with the use and understanding of technology. A person may have a brilliant mind and an outstanding intelligence, but without the five factor traits, a person cannot be one

TABLE 3.2

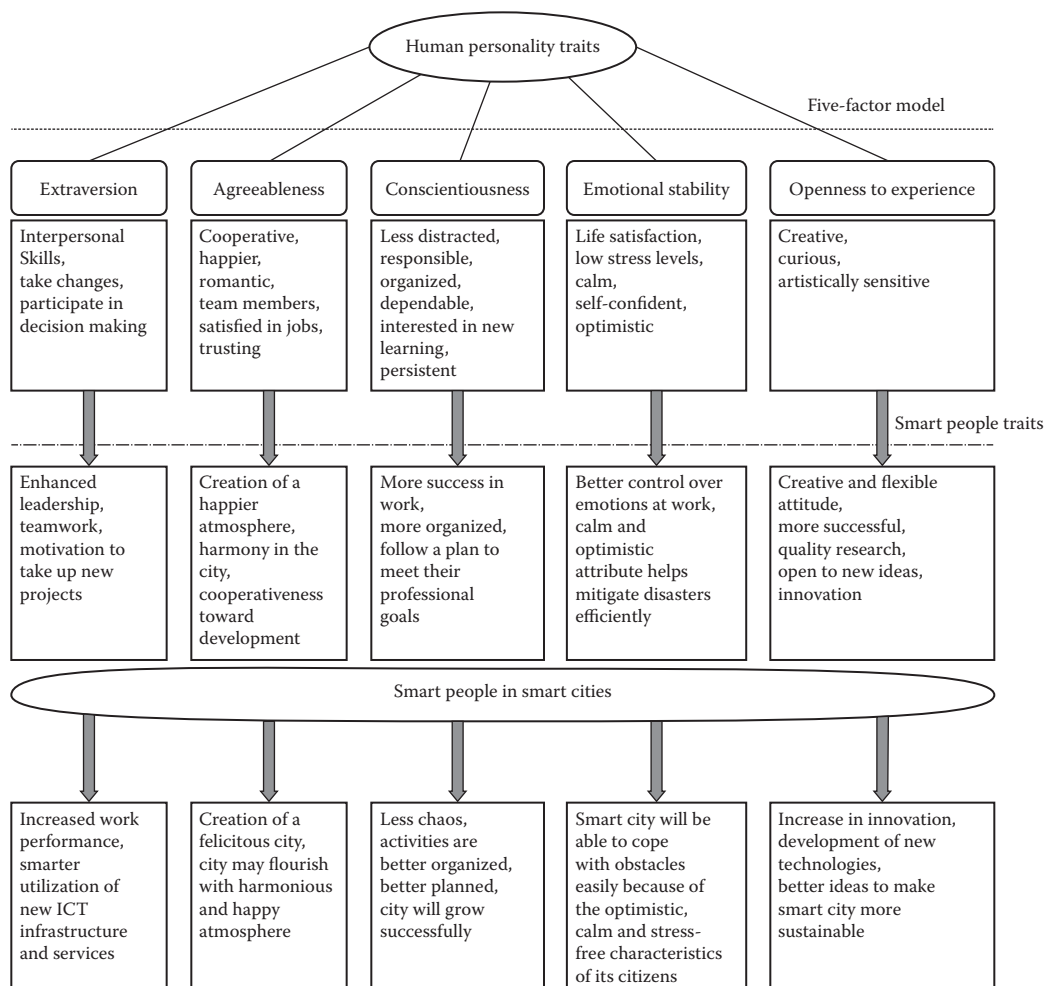
Relations among the Five-Factor Model and Smart People Traits

Personality Dimensions	People Performance	Quality in Smart People	Roles of People in Smart Cities
Extroversion	Better interpersonal skills, willing to take changes, participate actively in decision processes.	Enhanced leadership, better teamwork, motivated to take up new projects.	Increase in work performance, better and smarter utilization of new ICT infrastructure and services.
Agreeableness	Cooperative, warm, happy, romantic, organizational team members, more satisfied in jobs and trusting.	Creation of a happier and more positive atmosphere, harmony in the city, cooperativeness toward development.	Creation of a more felicitous city, city may flourish because of the harmonious, warm, and happy atmosphere.
Conscientiousness	Less distracted and more responsible, organized, dependable, interested in new learnings, and persistent.	Such people have more success in their work because they are more organized and always follow a plan to meet their professional goals.	Less chaos, processes and activities are better organized and always go according to plan, the city will grow successfully.
Emotional stability	Life satisfaction, job satisfaction, and low stress levels, calm, self-confident, optimistic.	Better control over emotions at work, calm and optimistic attribute helps mitigate disasters efficiently.	Smart city will be able to cope up with obstacles easily because of the optimistic, calm, and stress-free characteristic of its citizens.
Open to experience	Creative, curious, and artistically sensitive.	Creative and flexible attitude, more successful and higher-quality research, open to new ideas, innovation.	Increase in innovation, development of technologies, make smart cities more sustainable.

of the smart people. Unless people are smart enough within a city to make good use of ICT technology and create a healthy, happy city environment, the development and sustainability of the smart city seem questionable. The framework of smart people to develop a smart city is shown in [Figure 3.2](#).

3.5 Conclusion

The concept of the smart city varies from city to city and country to country. However, one thing that remains common in all definitions and characteristics of a smart city is citizen engagement. Hence, human factors and human development in smart cities is a very important domain to analyze. This chapter has attempted to provide a base to establish attributes that are important for people to be characterized as smart people. The big-five personality traits are powerful predictors of human behavior. Although they are mostly used in an organizational setup to recruit and hire people, these traits are equally beneficial in defining attributes of smart people. In a smart city, its citizens need to possess nearly all the attributes described in this study to become the smart people. Smart people can lead to the sustainable growth and development of smart cities.

**FIGURE 3.2**

Behavioral framework of smart people for the development of sustainable smart cities.

References

- Angelidou, M. 2015. Smart cities: A conjuncture of four forces. *Cities*, 47: 95–106.
- Castells, M. 1996. *The Rise of the Network Society*. Vol. 1 of *the Information Age: Economy, Society and Culture*. Oxford: Blackwell.
- Cattell, R. B. 1957. *Personality and Motivation Structure and Measurement*. Yonkers, NY: World Book Company.
- Costa, P. T., McCrae, R. R., and Dye, D. A. 1991. Facet scales for agreeableness and conscientiousness: A revision of the NEO Personality Inventory. *Personality and individual Differences*, 12, 9: 887–898.
- Downey, J., and McGuigan, J., Eds. 1999. *Technocities: The Culture and Political Economy of the Digital Revolution*. London: Sage Publications.
- Florida, R. 2005. *Cities and the Creative Class*. New York: Routledge.

- Giffinger, R., Fertner, C., Kramar, H., Kalasek, R., Pichler-Milanovic, N., and Meijers, E. 2007. Smart cities. Ranking of European medium-sized cities, Final Report, Centre of Regional Science, Vienna UT.
- Graham, S., and Marvin, S. 1999. Planning cybercities: Integrating telecommunications into urban planning. *Town Planning Review*, 70, 1: 89–114.
- Guthrie, G. M., and Bennett, A. B. 1971. Cultural differences in implicit personality theory. *International Journal of Psychology*, 6, 4: 305–312.
- Komninos, N. 2008. *Intelligent Cities and Globalization of Innovation Networks*. London; New York: Routledge.
- Kumar, T. V. 2015. E-governance for smart cities. In *E-governance for Smart Cities* (pp. 1–43). Singapore: Springer.
- Norman, W. T. 1963. Toward an adequate taxonomy of personality attributes: Replicated factor structure in peer nomination personality ratings. *Journal of Abnormal and Social Psychology*, 66, 6: 574.
- Shepard, M. 2011. *Sentient City: Ubiquitous Computing, Architecture, and the Future of Urban Space*. MIT Press.
- Tupes, E. C., and Christal, R. E. 1961. *Recurrent Personality Factors based on Trait Ratings*. No. ASD-TR-61-97. Lackland Air Force Base, TX: Personnel Research Laboratory.
- Zipf, A. Live cities and urban services: A multi-dimensional stress field between technology, innovation and society.